**FACULTY OF COMMERCE**

**DEPARTMENT OF INFORMATION SYSTEMS**

**INF 2007F: APPLYING DATABASE PRINCIPLES**

**Project Part 2 – SQL [50 Marks]**

**Due Date:** 23rd April 2026.

**Student Number:**

## Instructions

- **This is an individual assignment**.
- The assignment is based on the **African Safari Adventure** case study you did in Part 1.
- All submissions must be made electronically through AMATHUBA by 23h59 on the 23rd April 2026.
- Please note that no late submissions will be accepted.
- AI Use is prohibited.

**Objective:** This project builds directly on **Project Part I**, where you designed a conceptual EER model and mapped it to a normalised relational schema. You will now implement that schema in MySQL, populate it with realistic data, and write SQL queries to demonstrate your mastery of database principles.

### Relational Model (Normalised)

The provided **AfricanSafariDB** creates the tables based on the following model:

- **USER**(UserID, FirstName, LastName, Street, Suburb, City, Province, PostalCode, DOB)
- **SAFARI_PROVIDER**(ProviderID, LinkedIn)
- **CUSTOMER**(CustomerID, Contact)
- **PERFORMING_GROUP**(GroupID, GroupName, Bio, Price)
- **SAFARI_ADVENTURE**(AdventureID, Location, Street, Suburb, City, Province, PostalCode, Website, ProviderID)
- **PACKAGE**(PackageID, PackageName, Duration, Price, AdventureID)
- **BOOKING**(BookingID, BookingDateTime, PreferredDate, NumberOfPeople, CustomerID)
- **CULTURAL_ENT**(BookingID, EntertainmentType, Duration)
- **PHONE_NUMBER**(AdventureID, PhoneNumber)
- **GROUP_AVAILABILITY**(GroupID, AdventureID, Day)

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<u>**Questions**</u>

**Section A - Data Population (DML) [25 marks] (Write the insert statements in one file. Save it as DataPopulation.SQL)**

1. The User table has been populated for you. Using the users in this table, populate Safari Provider, Customers, and performing groups [9]

   - Safari Providers: Research **five (5) actual safari tour operators** currently operating in Southern Africa. Find their official LinkedIn company pages. Map these five providers to the first five users (UserID 1-5) in your USER table and populate the SafariProvider table.
   - Performing Groups: Research **five (5) real cultural performance groups** (e.g., traditional dance troupes, choirs, or fire dancers) based in Africa. Note their typical performance style and average booking fee. Populate the PerformingGroup table with this data.
   - Customers: Populate the CUSTOMER table for the remaining eight users (UserID 6-13) using realistic mobile contact numbers.

2. Create at least 3 adventure locations (Kruger National Park, Serengeti, etc). Use real addresses or landmarks for the street and suburb fields. Link these to your researched ProviderIDs. [3]

3. Add at least one contact number for each adventure location in the Phone_Number table. [2]

4. For each performing group, assign them to at least one AdventureID on specific days of the week (e.g., ‘Monday’, ‘Wednesday’) in the GroupAvailability table [3]

5. Using the Safari Providers you researched above, create at least 5 unique travel packages with realistic names, durations, and prices. The price of the packages should be between R1000 and R6000. Ensure these packages are correctly linked to your Safari Adventure locations. [3]

6. Generate at least 8 booking records. Distribute these among your customers and ensure that the BookingDateTime is realistic. [3]

7. For at least 4 of your bookings, add a record in the Cultural_Ent table to indicate that those customers requested cultural entertainment during their safari. [2]

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**Section B – Queries (SQL Practice) [25 marks] (All select queries should be in one file)**

1. Display the BookingID, BookingDateTime, and the City and Province of the customer who made the booking. [3]

2. Show each customer’s name, the package(s) they booked, and the performing group (if any). [4]

3. Find all packages that are priced between R2000 and R4000. List the PackageName, Duration, and Price, sorted from most expensive to cheapest. [4]

4. List the GroupName and the Location of the Safari Adventure where they are available to perform, based on the GroupAvailability table.

[4]

5. Count how many different packages are offered at each AdventureID. Display the AdventureID and the count, but only for locations that have more than one package.

[3]

6. List all customers in descending surname order who booked adventures with more than 4 people. [3]

7. List all PerformingGroups whose booking price is higher than the average price of all groups in the database. [4]

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